Updated Functional & Non-Functional Requirements Document (Version 3)

PART I: REVISION LOG & SUMMARY OF CHANGES (v2 → v3)

1. What Was Added and Why

- Authentication & Session Management (in detail): The Feature Tree treats Authentication & MFA and Session Management as two separate features, but v2 covered both in one sentence. v3 now spells out MFA rules, session timeout, secure logout, account lockout and control of multiple logins.
- Admin User Management: The Feature Tree has this as its own feature, while v2 only said the Admin can "manage users". v3 says exactly what the Admin can do: create and deactivate accounts, assign roles and assign advisors.
- Advisor Oversight: The Feature Tree shows this as a feature of its own. v3 gives advisors a clear view of pending approval requests and their assigned students, so they don't have to hunt for them.
- Academic History & CGPA Calculation: The Feature Tree lists Academic History and Exam & CGPA Calculation separately. v3 describes both properly: publishing results, calculating CGPA automatically, and showing the full academic history.
- Data Consistency: The System Performance diagram includes this, so v3 adds it as a requirement. Registration, seats, waitlists, grades and wallet payments must always stay accurate, with no seat given twice and no money spent twice.
- Wallet Security & Reconciliation: Because real money is involved, v3 adds clear handling for low balance, a PIN/OTP check for large payments, refunds for failed transactions, and an audit log.
- Admin & Advisor Reports: The Feature Tree groups Analytics & Dashboard together. v3 goes beyond the student dashboard and lets Admins and Advisors see registration trends and seat usage.

2. What Was Modified and Why

- Document Structure: The 22 sections in v2 are now grouped under the five Feature Tree branches (User & Auth Management, Academic & Registration, Campus Smart Wallet, Campus Life & Facilities, System & Operations). This way every Feature Tree item has a matching requirement.
- Course Deletion → Archiving: v2 let the Admin delete obsolete courses. Deleting a course that has registrations or grades would wipe out academic records, so v3 archives or deactivates it instead. A course can be deleted for good only if it has no history.
- Notifications Merged: v2 mentioned notifications in three different places. v3 brings them together into one requirement covering in-app, email and mobile push alerts.
- Grade Appeal Workflow: v2 only said the system "provides a formal process". v3 lays out the steps: the student appeals within a set period, a teacher or admin reviews it, approves or rejects with comments, the CGPA is updated, and every change is logged. The student can follow the status throughout.
- Messaging & Announcements: Merged into one feature, to match the Feature Tree.

- Non-Functional Requirements: Vague words like "rapidly" and "high" are replaced with numbers we can actually test, such as response time, uptime, number of users and WCAG level.
- Teacher Items Regrouped: Teacher assignment and class roster were sitting under Course Management in v2. They now sit under Teacher Requirements, where they belong.

3. What Was Removed and Why
   - Duplicate Statements: Some rules were repeated across v2 sections (Student Requirements, Course Registration, Registration Rules, Seat Availability). They are merged so each rule appears only once.
   - Attendance Tracking (still excluded): This stays out of scope. It needs daily class-level records, which belong in a Learning Management System (LMS).
   - Health & Support Services (still excluded): This stays out of scope because of medical privacy rules and specialised healthcare workflows.

PART II: FINAL REQUIREMENTS SPECIFICATION

BRANCH A: USER & AUTH MANAGEMENT

1. User Roles, Permissions & Authentication
   - Student: Can register for courses, manage their account, view academic records and use the wallet.
   - Teacher: Can manage assigned courses, see class rosters, upload course information and submit grades.
   - Advisor: Can approve or reject student course registrations, keep an eye on assigned students and give guidance.
   - Administrator: Can manage users, courses, system settings, registration periods and campus services.
   - Authentication & MFA: Users log in securely with Multi-Factor Authentication (required for Teachers, Advisors and Administrators, optional for Students) and can reset a forgotten password.
   - Session Management: Sessions end automatically after a period of inactivity. Users can log out securely, accounts lock after repeated failed logins, and multiple logins on one account are controlled.
   - Role-Based Access Control (RBAC): Each user can only use the functions that belong to their role.
2. Account Management & Admin User Management
   - Account Management: Students can create an account and update their personal information.
   - User Administration: Admins can create, update, deactivate and reactivate student, teacher, advisor and administrator accounts.
   - Role & Advisor Assignment: Admins can assign roles to users and assign an advisor to each student.
3. Registration Period Administration
   - Registration Period Management: Admins can set, open and close the course registration window for each term.

- Course Withdrawal Deadline: Admins can set the last date for withdrawing from courses each term.
- System Rule Management: Admins can manage the global registration rules and limits, such as credit limits and waitlist limits.

4. Academic Advisor Requirements (Advisor Oversight)
   - Student Oversight: Advisors can see the academic information and registration status of their assigned students.
   - Registration Review: Advisors can approve or reject registration requests, and give a reason when they reject one.
   - Comments & Guidance: Advisors can leave comments and suggest courses.
   - Advisor Overview: Each advisor gets a simple list of pending requests and assigned students.

BRANCH B: ACADEMIC & REGISTRATION

5. Course Management Requirements
   - Course Creation: Admins can add new courses.
   - Course Maintenance: Admins can update a course's description, credits, schedule, capacity and rules.
   - Course Archiving: Admins can archive courses that are no longer offered. A course can be permanently deleted only if it has no registration or grade history.
   - Capacity Management: The system keeps track of seat limits and prerequisites for each course.
6. Teacher Requirements
   - Course Roster: Teachers can see their assigned courses, class details and the list of enrolled students.
   - Course Information: Teachers can upload and update the course syllabus and other course information.
   - Results Upload: Teachers can upload exam results for the courses they teach.
7. Course Information Requirements

The system provides the following course attributes:

   - Fields: Course Name, Course Code, Credit Hours, Teacher, Prerequisites, Course Description, Available Seats, Class Schedule, Classroom Location.

8. Student Information Requirements

The system maintains the following student profile fields:

   - Fields: Student ID, Name, Department, Program, Completed Courses, Current Courses, Academic Results, Credits Completed, Remaining Credits.

9. Course Registration Requirements
   - Search & Selection: Students can search for courses by name, code, department, teacher or schedule, and pick the sections they want.
   - Prerequisite Checking: The system checks prerequisites automatically and won't allow a registration if they aren't met.
   - Conflict Prevention: The system stops students from registering for classes that clash in time, or for the same course twice.
   - Credit Limits: The system enforces the minimum and maximum credit limits.

- Registration Window: Students can register only while the registration period set by the Admin is open.
- Registration Confirmation: After submitting, students see a confirmation and a summary of their registration.

10. Waitlist & Capacity Requirements
    - Seat Availability: The system checks in real time whether seats are still available.
    - Waitlist Queue: If a section is full, the student is placed on a waitlist, first come, first served.
    - Waitlist Promotion: When a seat opens up, it is offered to the first eligible student on the waitlist, who is notified straight away.
    - Waitlist Status: Students can see their place in the queue and leave the waitlist whenever they like.
11. Course Drop & Withdrawal Requirements
    - Course Drop: Students can drop a course any time before the Course Withdrawal Deadline.
    - Deadline Enforcement: After the deadline, the system blocks the drop and tells the student why.
    - Seat Release: When a student drops a course, the seat is freed and the waitlist process starts.
12. Timetable Requirements
    - Schedule Generation: The system builds a timetable automatically from the courses a student has registered for.
    - Display Details: Students can see their weekly classes with times and classroom locations.
    - Conflict Detection: The system spots overlapping classes and lets the student know when the timetable changes.
13. Recommendation Requirements
    - Academic Analysis: The system looks at a student's academic history and the courses they have already completed.
    - Interest & Rules Matching: It also takes into account the student's interests, prerequisites and degree requirements.
    - Recommendation Display: Students can see a personalised list of recommended courses.
14. Exam, Academic History, CGPA & Grade Appeal Requirements • Exam Schedule: Students can view published exam schedules.
    - Result Publication: Teachers or Admins upload results, and students can see them once they are published.
    - CGPA Calculation: The system calculates GPA and CGPA automatically and keeps the academic history up to date.
    - Academic History: Students can view their academic history, GPA/CGPA and progress towards their degree.
    - Grade Appeal / Correction Workflow: Students can appeal a grade within a set appeal period. A teacher or admin reviews it and approves or rejects it with comments. An approved correction updates the result and CGPA, notifies the student and is recorded in an audit trail.

- Appeal Status Tracking: Students can follow the status of their appeal: submitted, under review, approved or rejected.

BRANCH C: CAMPUS SMART WALLET

15. Campus Smart Wallet & Payment Requirements
    - Pay for Food: Students can pay at campus canteens and cafeterias by tapping their card or using the app.
    - Pay for Campus Bus: Students can pay bus fares straight from their wallet.
    - Pay for Electricity: Students can top up their dorm room's electricity from the wallet.
    - Wallet Balance Check: Students can see how much money is left in their wallet, updated in real time.
    - Transaction History: Students can see every payment and top-up, and filter them by date and type.
    - Recharge / Top-up: Students can add money to the wallet through an online payment gateway and get a confirmation.
    - Insufficient Balance Handling: If the balance is too low, the payment is declined and the student is told why.
    - Payment Confirmation: Large payments need an extra check, such as a PIN or OTP.
    - Failed Transaction Reconciliation: Failed or duplicate transactions are sorted out automatically, and refunds show up in the transaction history.
    - Audit Log: Every wallet transaction is saved in a tamper-proof log that authorised administrators can check.

BRANCH D: CAMPUS LIFE & FACILITIES

16. Dormitory & Accommodation
    - Student Features: Students can apply for a dormitory room and see which rooms are available and where their application stands.
    - Admin Features: Admins can manage room allocations and accommodation status.
17. Library Requirements
    - Catalog & Borrowing: Students can search for books, see what they have borrowed and keep track of due dates.
    - Reservations & Alerts: Students can reserve books and get a notice when a book is overdue.
18. Campus Events
    - Events: Students can browse and sign up for campus events. Admins can create, update and cancel events.
19. Messaging & Announcements
    - Messaging: Students, teachers and advisors can message each other directly.
    - Announcements: Admins can post announcements to everyone or to a specific group.

BRANCH E: SYSTEM & OPERATIONS

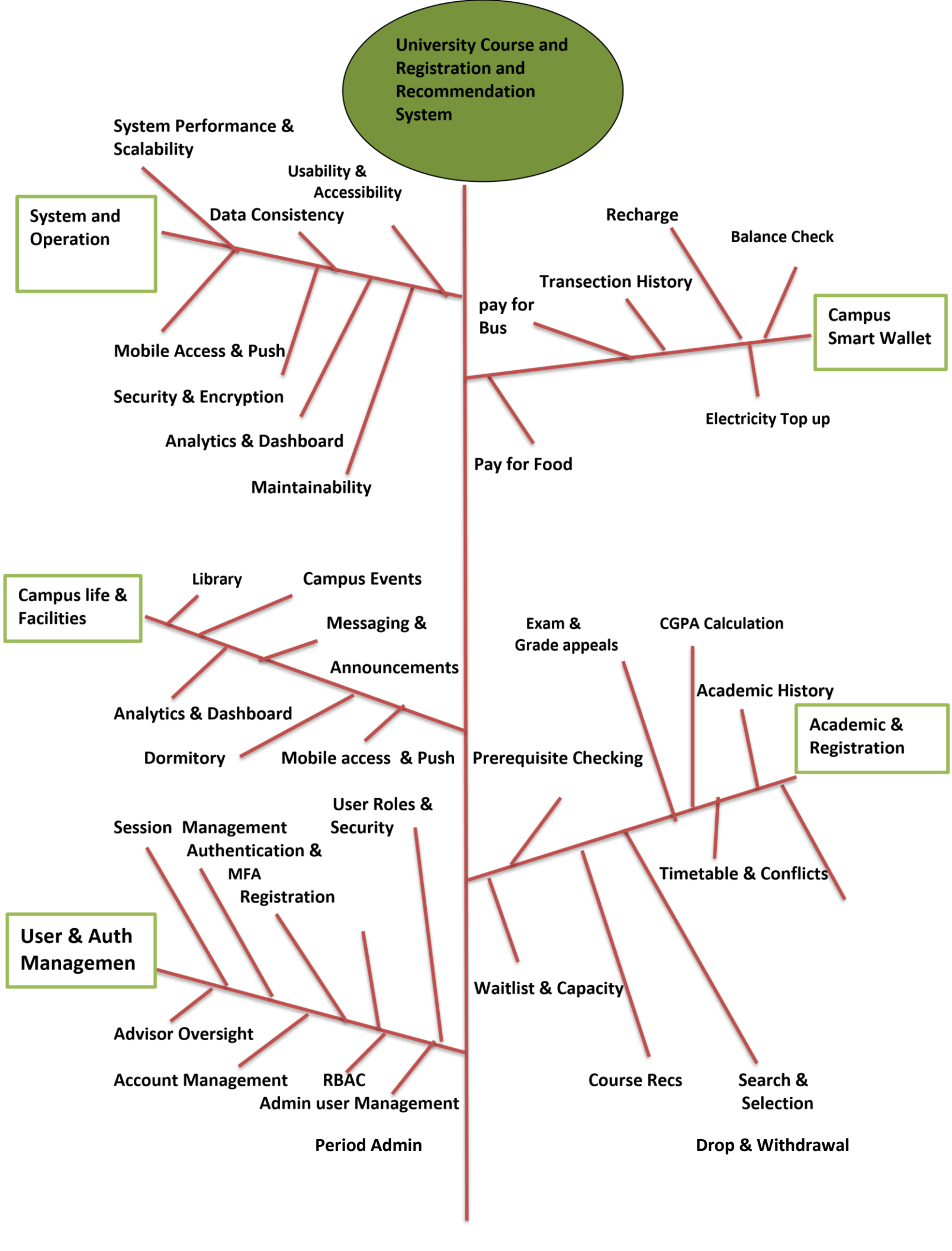
20. Notification Requirements
    - Channels: Notifications reach users through in-app alerts, email and mobile push.
    - Triggers: Users are notified about registration deadlines, approvals and rejections, waitlist updates, course changes, timetable changes, new recommendations, grade appeal status, wallet transactions and academic announcements.

21. Mobile Access & Push
    - Mobile Access: Students can use registration and academic features from their phones.
    - Mobile Push: Alerts and notifications are sent to students' registered mobile devices.
22. Analytics & Dashboard
    - Academic Insights: The system looks at how a student is performing and shares useful insights.
    - Student Dashboard: Students get a personal dashboard with their registration status, timetable, results and alerts.
    - Admin & Advisor Reports: Admins and Advisors can generate reports on registration trends, seat usage and student progress.
23. System Outputs
    - Outputs: Registration Confirmations, Student Timetables, Course Recommendation Lists, Registration Summaries, Academic Summaries, Notifications and System Reports.

PART III: NON-FUNCTIONAL REQUIREMENTS

- Performance: Everyday pages and searches should respond within 2 seconds, and a registration request should be processed within 3 seconds under normal load. The system must cope with the rush when registration opens, with a target of at least 5,000 users at the same time (to be confirmed against the real student intake).
- Security & Encryption: Data is encrypted while travelling (TLS) and while stored. Passwords are stored hashed and salted. Sessions are secure and access follows RBAC. Logins, grade changes, admin actions and financial transactions are all logged. Payments go through a compliant gateway, and full card details are never stored.
- Reliability: No data is lost during transactions. Backups run automatically at least once a day, and recovery targets are set by the university.
- Data Consistency: Registrations, seats, waitlists, grade changes and wallet payments are processed as complete, all-or-nothing steps. Even with many users at once, no seat is given out twice and no balance is spent twice.
- Availability: The system should be up 99.9% of the time while registration is open.
- Usability: The interface is easy to use on both desktop and mobile, so a new student can register without any training.
- Scalability: The infrastructure can grow with the student intake, without a redesign.
- Maintainability: The system is built in modules, so updates can go live without taking it offline.
- Accessibility: The system follows WCAG 2.1 Level AA so people with different abilities can use it.
- Compatibility: It works on current versions of the major browsers and on Android and iOS devices.

✞ Feature Tree:


University Course and Registration and Recommendation System
System and Operation
System Performance & Scalability
Usability & Accessibility
Data Consistency
Mobile Access & Push
Security & Encryption
Analytics & Dashboard
Maintainability
Campus Smart Wallet
Recharge
Balance Check
Transection History
pay for Bus
Electricity Top up
Pay for Food
Campus life & Facilities
Library
Campus Events
Messaging & Announcements
Analytics & Dashboard
Dormitory
Mobile access & Push
Academic & Registration
Exam & Grade appeals
CGPA Calculation
Academic History
Prerequisite Checking
Timetable & Conflicts
Waitlist & Capacity
Course Recs
Search & Selection
Drop & Withdrawal
User & Auth Managemen
User Roles & Security
Session Management
Authentication & MFA
Registration
Advisor Oversight
Account Management
RBAC
Admin user Management
Period Admin